Near-isometric chain lifting in reflexive, dual, and bidual Banach lattices

A partial result for arXiv:2103.08170

11 August 2026

Abstract

The open optimal-constant problem for projective free Banach lattices over lattices remains unresolved in the unrestricted category. We prove three sharp relative results. For every linearly ordered set L, $\mathrm{FBL}\langle L\rangle$ has 1^+-controlled lifts through quotients of reflexive Banach lattices and through weak-star normal quotients of dual Banach lattices. Every quotient of an arbitrary Banach lattice admits the same lift after passing to the bidual. If L is countable, quotients of $C(K)$-spaces admit an exact norm-preserving lift. The proofs isolate the remaining obstruction: finite 1^+-lifts glue by compactness, but a general Banach lattice supplies neither compact bounded fibers nor a lattice-preserving descent from its bidual.

1 Problem and notation

For a lattice L, write $P_L = \mathrm{FBL}\langle L\rangle$, and denote its canonical generators by δ_ℓ, $\ell \in L$. Its universal property says that every bounded lattice homomorphism $s : L \to X$ extends uniquely to a Banach-lattice homomorphism $\widehat{s} : P_L \to X$, with

$$\|\widehat{s}\| = \sup_{\ell\in L} \|s(\ell)\|.$$

For linearly ordered L, the source paper proves that P_L is projective precisely for the countable bounded chains, but obtains only a 2^+ upper bound and asks for the optimal constants [1]. Finite lattices are known to generate 1^+-projective free Banach lattices [2].

For a closed ideal J in a Banach lattice X, let $Q : X \to X/J$ be the quotient map. We write $\kappa_X : X \to X^{**}$ for the canonical embedding.

2 Compact gluing

Lemma 1 (finite-lift compactness)**.** *Let L be a linearly ordered set, let Z be a Banach lattice whose closed ball RB_Z is compact for a Hausdorff topology τ, and suppose the positive cone of Z is τ-closed. Let $q : Z \to Y$ be a lattice quotient such that every fiber*

$$K_\ell = \{z \in RB_Z : qz = t_\ell\} \qquad (\ell \in L)$$

is nonempty and τ-closed. If every finite subchain $F \subset L$ has representatives $z_\ell \in K_\ell$, $\ell \in F$, satisfying $z_\ell \le z_m$ whenever $\ell \le m$, then the entire family $(t_\ell)_{\ell\in L}$ has such representatives in RB_Z.

Proof. Each K_ℓ is compact, so $\prod_{\ell\in L} K_\ell$ is compact. For $\ell \le m$, the condition $z_\ell \le z_m$ is closed because the positive cone is closed. These closed conditions have the finite intersection property by hypothesis. Their total intersection is therefore nonempty. $\square$

Theorem 1 (relative 1^+-projectivity and bidual lifting)**.** *Let L be any linearly ordered set, $P_L = \mathrm{FBL}\langle L\rangle$, and $\varepsilon > 0$.*

1. *If X is reflexive, then every lattice homomorphism*

$$T : P_L \longrightarrow X/J$$

has a lift $\widetilde{T} : P_L \to X$ satisfying

$$Q\widetilde{T} = T, \qquad \|\widetilde{T}\| \le (1+\varepsilon)\|T\|.$$

2. *The same conclusion holds when $X = E^*$ is a dual Banach lattice with its canonical weak-star closed cone and J is a weak-star closed ideal.*

3. *For arbitrary X and J, there is a bidual lift $\widetilde{T} : P_L \to X^{**}$ such that*

$$Q^{**}\widetilde{T} = \kappa_{X/J}T, \qquad \|\widetilde{T}\| \le (1+\varepsilon)\|T\|.$$

Consequently the infimal lifting constant in each of the first two relative categories, and in the bidual sense of the third, is exactly 1.

Proof. Put $C = \|T\|$; the case $C = 0$ is immediate. For each $\ell \in L$, set $t_\ell = T\delta_\ell$. If $F \subset L$ is finite, then F is a finite lattice. The map $t|_F : F \to X/J$ induces a lattice homomorphism

$$T_F : P_F \longrightarrow X/J, \qquad \|T_F\| = \max_{\ell\in F} \|t_\ell\| \le C.$$

The 1^+-projectivity theorem for finite lattices gives a lift of T_F of norm at most $(1+\varepsilon)C$. Its values on the generators are representatives of the t_ℓ's, are ordered along F, and lie in the ball of radius $R = (1+\varepsilon)C$.

If X is reflexive, apply Lemma 1 to the weak topology on RB_X. Quotient fibers are weakly closed, and the positive cone is weakly closed. We obtain an order-preserving map $s : L \to RB_X$ with $Qs(\ell) = t_\ell$. Because L is a chain, order preservation is equivalent to preservation of its finite meets and joins. The universal property gives $\widetilde{T} = \widehat{s}$, with $\|\widetilde{T}\| \le R$, and equality on generators gives $Q\widetilde{T} = T$.

For $X = E^*$, use the weak-star topology. Its norm ball is weak-star compact and its positive cone is weak-star closed. A weak-star closed ideal J makes X/J a dual quotient and Q weak-star continuous, so its fibers are weak-star closed. The same argument applies.

For arbitrary X, work in X^{**} with its weak-star topology and replace each finite lift $x_\ell \in X$ by $\kappa_X x_\ell$. Define

$$K_\ell = \{x^{**} \in RB_{X^{**}} : Q^{**}x^{**} = \kappa_{X/J}t_\ell\}.$$

These are nonempty weak-star compact sets; the cone of X^{**} is weak-star closed, and Q^{**} is weak-star continuous. Compact gluing produces an ordered $s : L \to X^{**}$, and the universal property gives the asserted bidual lift.

Finally, every lift has norm at least $\|T\|$, since the quotient and canonical embedding are contractive. Allowing arbitrary $\varepsilon > 0$ proves the last assertion. $\square$

3 Exact lifting through $C(K)$-quotients

The compactness proof uses an arbitrarily small norm loss inherited from the finite-lattice theorem. For countable chains and AM-space quotients, one can remove even that loss.

Lemma 2 (relative ordered Tietze extension)**.** *Let F be closed in a compact Hausdorff space K. If $a, b \in C(K)$, $a \leq b$, and $f \in C(F)$ satisfies $a|_F \leq f \leq b|_F$, then there exists $g \in C(K)$ such that*

$$g|_F = f, \qquad a \leq g \leq b.$$

Proof. By the ordinary Tietze theorem, choose $h \in C(K)$ with $h|_F = f$. Then

$$g = (h \vee a) \wedge b$$

is continuous and lies between a and b. On F, the assumed inequalities give $(f \vee a|_F) \wedge b|_F = f$, so $g|_F = f$. □

Theorem 2 (norm-one $C(K)$-quotient lifting)**.** *Let L be a countable linearly ordered set, F a closed subset of a compact Hausdorff space K, and $R_F : C(K) \to C(F)$ the restriction quotient. Every lattice homomorphism*

$$T : P_L \longrightarrow C(F)$$

has a lattice-homomorphic lift $\widetilde{T} : P_L \to C(K)$ with

$$R_F\widetilde{T} = T, \qquad \|\widetilde{T}\| = \|T\|.$$

Proof. Set $C = \|T\|$, $t_\ell = T\delta_\ell$, and enumerate $L = \{\ell_1, \ell_2, \ldots\}$. We recursively construct $g_n \in C(K)$ such that

$$g_n|_F = t_{\ell_n}, \qquad \|g_n\|_\infty \leq C,$$

and $g_i \leq g_j$ whenever $i, j \leq n$ and $\ell_i \leq \ell_j$.

Suppose $g_1, \ldots, g_{n-1}$ are defined. Let

$$a_n = \max\bigl(\{-C\mathbf{1}\} \cup \{g_j : j < n,\ \ell_j < \ell_n\}\bigr), \quad b_n = \min\bigl(\{C\mathbf{1}\} \cup \{g_j : j < n,\ \ell_n < \ell_j\}\bigr).$$

The previous order relations give $a_n \leq b_n$. On F, the fact that $\ell \mapsto t_\ell$ is order preserving gives

$$a_n|_F \leq t_{\ell_n} \leq b_n|_F.$$

Lemma 2 supplies g_n between these two bounds and extending t_{ℓ_n}. This completes the recursion.

The map $s(\ell_n) = g_n$ is a bounded lattice homomorphism from the chain L to $C(K)$, with rank at most C. Its universal extension $\widetilde{T} = \widehat{s}$ has norm at most C and restricts to T. The reverse inequality follows from contractivity of R_F. □

4 Why this does not settle the unrestricted problem

Theorem 1(3) shows that the finite-lattice estimates already contain a coherent global lift; it naturally lives in X^{**}. A general principle of local reflexivity cannot finish the proof: it is linear and finite-dimensional, whereas the required descent must simultaneously preserve the lattice order for all generators and commute exactly with Q. Nor does diagonal convexification solve the problem. Convex combinations preserve the order relations of a chain and the quotient equations, but a bounded sequence of finite lifts need not have a weakly convergent subnet in a nonreflexive space.

The recursive proof of Theorem 2 uses the special relative insertion property of $C(K)$. For a general Banach-lattice quotient, clipping a lift between ordered endpoint lifts gives the familiar factor 2, since

$$\big\||a| \vee |b|\big\| \leq \|a\| + \|b\|,$$

and no near-isometric simultaneous insertion theorem is available. Thus the unrestricted question for countable bounded chains reduces sharply to whether the bidual ordered lift can always be descended, with arbitrarily small norm loss, through an arbitrary Banach-lattice quotient.

References

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[2] A. Avilés and J. D. Rodríguez Abellán, *Projectivity of the free Banach lattice generated by a lattice*, Arch. Math. 113 (2019), 515–524, https://arxiv.org/abs/1903.01191.